Project Specification
StorePulse: Sales and Customer Analytics Dashboard
Version 1.0
A small data analytics project based on simulated retail sales data, focused on SQL analysis and data visualization.

Objective
- The objective of the project is to design a simple analytics pipeline that stores structured retail data, extracts useful business metrics and presents them in clear visual form.
- The final result should help answer basic business questions such as which products sell best, how sales change over time, and which customer segments are most active.

Project Description
- The project will use a synthetic dataset that represents a small online or offline store.
- The dataset will include customers, products, orders and order items.
- Data will be stored in a relational database and analyzed with SQL queries and Python scripts.
- The output will be a set of charts and a short insight summary prepared in a report format.

Core Features
- Synthetic data generation for customers, products, orders and order items.
- SQL database schema with basic relational structure.
- Data loading and querying with SQL.
- Aggregation of key metrics such as total revenue, average order value, monthly sales, top products and top categories.
- Python-based data analysis using pandas.
- Visualizations such as line charts, bar charts and histograms.
- A short final report with main insights and observations.

Data Model
- Customers: customer_id, full_name, city, registration_date
- Products: product_id, product_name, category, price
- Orders: order_id, customer_id, order_date, total_amount
- Order_Items: order_item_id, order_id, product_id, quantity, item_price

Functional Requirements
- Generate a realistic synthetic dataset.
- Load the data into a relational database.
- Run SQL queries to calculate business metrics.
- Produce at least four visualizations.
- Summarize findings in a short report.

Non-Functional Requirements
- The code should be readable and well structured.
- The project should run locally without complex setup.
- The dataset should be reproducible where possible.
- The final solution should be lightweight and easy to demonstrate.

Technologies
- SQL for schema design and data queries.

- Python for analysis, data processing and visualization.
- pandas for tabular data handling.
- matplotlib for charts.
- sqlite3 or PostgreSQL for database storage.
- C++ may be used optionally for a small helper utility, such as data generation, but it is not required.

Deliverables

- Database schema script.
- Data generation script or source file.
- SQL queries for analytics.
- Python analysis and visualization scripts.
- Charts and a short final report.
- README file with setup and run instructions.

Acceptance Criteria

- The dataset is generated successfully and loaded into the database.
- The SQL queries return correct aggregated results.
- The Python analysis produces meaningful insights.
- The visualizations are clear and readable.
- The final report explains the main findings in a concise way.